Static Analysis for Bounds Check Elision in the Naga WGSL Compiler

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Paper #55

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Proving array indices in bounds removes 102 of the 186 bounds clamps in vello's GPU shaders.

On an Intel iGPU that wins back the full cost of the checks: **+7.0%**. On NVIDIA the checks cost only 1.3% in total, so the gain is small: **+0.6%**.

+7.01% / +0.59%: pre-registered suite geomeans over 32 vello scenes each, 95% CI +3.69 to +10.42% and +0.15 to +1.03%. Total check cost (checks on vs. off) is exploratory.

code, data and the source of every number

How it works: one proof, and a tree reduction loses all four clamps

The loop (reduce_tree)

```
@workgroup_size(256)
var<workgroup> s: array<u32, 256>;

for (var stride = 128u; stride > 0u;
     stride = stride >> 1u) {
  if (lid < stride) {
    s[lid] = s[lid] + s[lid + stride];
  }
  workgroupBarrier();
}
```

What the analysis derives

1. `lid` ∈ [0, 255]: the workgroup size
2. `stride` ∈ [1, 128]: its only update is >> 1
3. `lid < stride` ⇒ `lid` ∈ [0, 127]
4. `lid + stride` ∈ [1, 255] fits s, **so no clamp is needed**

Emitted SPIR-V (Restrict)

```
~~UMin %lid %uint_255~~
~~UMin %lid %uint_255~~
~~UMin %lid_plus_stride %uint_255~~
~~UMin %lid %uint_255~~
```

4 → 0 clamps. The binary is byte-identical (sha256) to the build with all checks off.

Shown with wgpu's Vulkan settings; SPIR-V ids renamed for reading. Artifact: runs-v4, wgpu fork 19ffb4e.

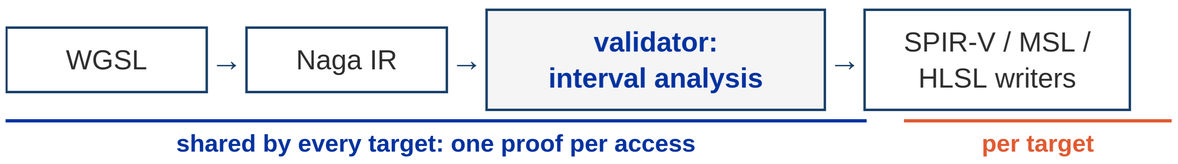
1 Today, wgpu clamps array indices

WGSL forbids out-of-bounds memory access. On Vulkan, Metal and D3D12, wgpu rewrites `s[i]` as `s[min(i, 255)]`.

	arrays & vectors	storage buffers
Vulkan	clamp	hardware (robustBufferAccess2)
Metal	clamp	clamp
D3D12	clamp	clamp
GLES	none	none

A browser must keep these checks. Native apps can turn them off (vello does).

2 How a fact is proved



Every integer carries an **interval**. Branches narrow it; a loop variable matches a known **recurrence** or becomes unknown. A proved access loses its clamp.

Branch guard

```
var i = data[g.x];
if (i < 8u) { … lut[i] … }
```

i ∈ [0, 7] in the branch

Bit mask

```
lut[data[g.x] & 7u]
```

x & 7 ∈ [0, 7]

Loop recurrence

```
for (var k = 128u; k > 0u;
     k = k >> 1u)
```

k ∈ [1, 128] in the body

Relation a ≥ b

```
if (lid >= off) {
  t = s[lid - off]; }
```

lid − off ∈ [0, 254]

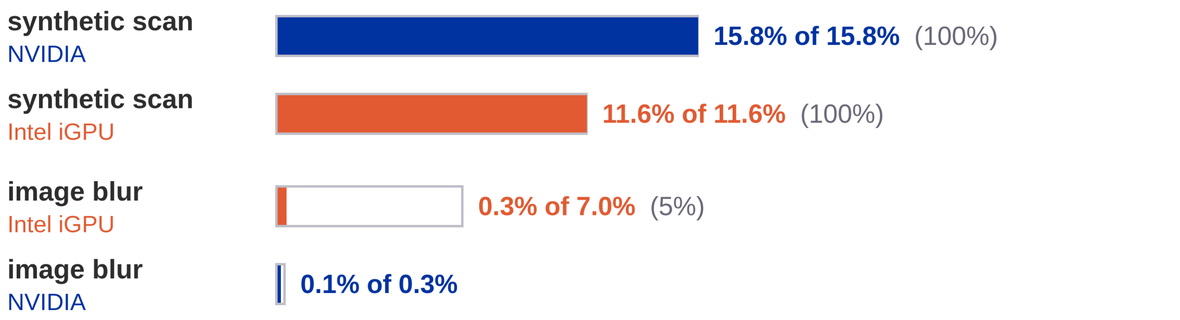
Not proved, so the clamp stays: runtime-sized arrays, bounds like `i < arrayLength(&a)`, facts across calls, guards on `let` values.

3 Why trust a removed check?

- **Tests:** 95 hand-written shaders where a check must stay or must go; interval arithmetic checked against exhaustive enumeration.
- **On GPU:** 8 of 8 differential runs give byte-identical output with and without elision.
- **Audit:** we found 3 unsound loop/branch shapes and fixed each with a regression test. No measured binary changed.
- **Not yet covered:** shapes beyond these tests, and `discard` on Metal/D3D12. Next: random-program differential testing.

4 Check cost, and what elision wins back

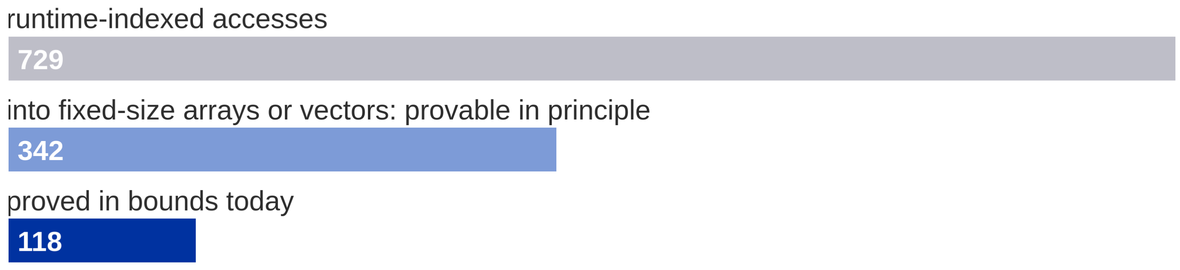
Outline: slowdown from all bounds checks (checks on vs. off). **Fill**: the part elision removes.



Where checks are expensive and provable, elision wins back all of their cost, on both GPUs. On NVIDIA, vello's checks are nearly free. Blur's checks cost 7% on Intel but are mostly not provable (section 5).

Exploratory: raw medians, not pre-registered. Every synthetic kernel with a measurable check cost recovers 95–100% (5 on NVIDIA, 3 on Intel). Why checks cost more on Intel is not yet known.

5 Coverage in 9 open-source projects



364 accesses (49.9%) index runtime-sized arrays: the next step is symbolic lengths.

6 Compared with Tint (Chrome)

	this work	Tint
facts from any `if` guard	yes	loop exits only
relation a ≥ b for a − b	yes	no
shrinking / geometric loops	yes	±1 steps only
runtime-sized arrays	no	no
shipped	no (fork)	default-on

From reading Tint's IntegerRangeAnalysis source, not measured. Classic CPU work: ABCD, Go `prove`, LLVM IRCE.